

Danil Karpov

Research Engineer | Machine Learning

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ML Background

Machine Learning Fundamentals · Statistical Modelling · Computer Vision · Representation Learning · Bayesian Optimisation · Data Selection and Sampling · Explainable AI

Education

B.Sc. in AI Engineering Sep 2024 — Present
ITMO University — Artificial Intelligence Technologies Faculty *[link]* St. Petersburg, Russia

- Current GPA: **5.0 / 5.0** (4 consecutive semesters)

Yandex School of Data Analysis (SDA) Sep 2026 — Present
Data Science Track; expected completion 2028

Selected coursework (ITMO & YSDA)

Work Experience

ML Researcher Feb 2025 — Present
Yandex · Kinopoisk St. Petersburg, Russia

- **Video-quality metric.** Researched and developed a perceptual quality metric from noisy human assessments. Devised the data-collection strategy and formulated a **probabilistic model** for standardising subjective ratings. Proposed and investigated an approach combining a larger predictive model, dataset expansion and **distillation** into a metric whose predicted quality **decreases monotonically with CRF**, enabling production search for a compression setting that meets a target quality. Evaluated alternative monotonic model families and deployed the resulting metric.
- **Dataset construction.** Redesigned my earlier sampling approach around **VAE representations**, addressing its limitations and scaling to millions of video clips. Investigated latent representations using a 2.5-million-clip data sample. Designed the method for **reuse across tasks** by separating representation learning from task-specific selection objectives; it has already supported the construction of two fundamentally different datasets. Manual review showed less repetition and more diverse atypical clips than with earlier approaches.
- **ML-guided video encoding.** Proposed and developed a multi-stage ML approach to video encoding, using predicted quality and bitrate curves to guide iterative refinement. Combined with optimised quality-metric computation, the approach **approximately halved estimated CPU cost** relative to the existing pipeline in offline evaluation on 1000 AVC/1080p clips.
- **Encoding preset optimisation.** Used Bayesian optimisation to investigate x264/x265 presets. Identified a mismatch between objective quality scores and visible artefacts in animated content, and refined candidate settings through subjective validation. The HEVC campaign achieved **∼14% lower bitrate** than the existing production baseline at unchanged resolution and encoding speed; the resulting preset was deployed in March 2026.
- **Core team tooling.** Developed reusable tools for scene segmentation and dataset assembly. Generalised the team's Bayesian optimiser into a configurable, resumable tool used across encoding-preset experiments.

Hackathons & Competitions

AI Challenge — **1st place** / 40 teams ($10 000 prize) [link] 2023
ECG signal processing, researched signal classifier architectures, implemented a bagging pipeline of 1D CNNs.

AI Challenge — **3rd place** / 30 teams ($6 000 prize) [link] 2024
YOLO fine-tuning, continuous data annotation, and various custom optimizations to maximize the score.

AIDAO — **4th place** / 50 teams [link] 2025
Multi-view car image classification to verify compliance with taxi standards via training and fine-tuning CNNs, implemented Docker instance.

AI NTO — **6th place** / 40 teams [link] 2024
Developed a genetic algorithm operating on molecular structures.

Public Talks & Teaching

Course on Explainable AI (XAI) Feb 2026 — Present
St. Petersburg State University (SPbU)

- Developed and delivered a course covering model-agnostic methods, uncertainty, robustness & attacks, and trustworthy LLMs. Course Materials

Extra Activities

SMILES 2026 — Summer School of Machine Learning **[link]** July 2026
Skoltech & Nanjing University Suzhou, China

- Completed a team research project with a final defence, with all lectures and presentations in English. Coverage: LLMs & generative AI, multi-agent systems, and safe & reliable ML.

SHAD Agents Week Intensive April 2026
Yandex School of Data Analysis

- Completed a focused program on LLM-based autonomous agents, tool use, and agentic workflows.

Languages

Russian Native
English Upper-Intermediate (B2)